

Stage 1 Task 2 Report

Data Science Project, Group 8

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Team Section - COVID-19 Data

COVID-19 Dataset Analysis

The COVID-19 dataset consists of three separate datasets: Cases, Deaths, and County Population. After aggregating these datasets, we are left with a table that provides information about COVID-19 cases and deaths on a by-county basis for the entire United States. As a result, we can easily incorporate any other information that is collected on the county level for further analysis. The variable dictionary for our aggregated dataset, before adding any enrichment data, is as follows:

COVID-19 Dataset Variable Dictionary

Aggregated COVID-19 Dataset				
Name	Definition	Data Type	Possible Values	Required
countyFIPS	FIPS Code for the county. These are unique to each county.	int64 (integer)	0, 1001, 56045	Yes
County Name	Name of the County. These are not necessarily unique.	object (string)	Weston County, Yakutat Borough, Yukon-Koyukuk Census Area	Yes
State	Two Letter State Abbreviation.	object (string)	AL, NC, WY	Yes
StateFIPS	FIPS Code for the state.	int64 (integer)	1, 25, 56	Yes
County Population	Population of the county.	int64 (integer)	0, 24686, 10039107	Yes
yyyy-mm-dd_cases	Cumulative number of confirmed cases in this county as of this date.	int64 (integer)	0, 8149, 70521	Yes
yyyy-mm-dd_deaths	Cumulative number of COVID related deaths in this county as of this date.	int64 (integer)	0, 8149, 70521	Yes
NOTE: There is a column for Cases and a column for Deaths for each day from 1/22/2020 to 7/23/2023				

The dataset uses the Federal Information Processing System (FIPS) Codes to uniquely identify both counties and states. This is used because county names are not necessarily unique, and may be repeated between states. For example, There is a Randolph County in North Carolina, but also in Alabama, Arkansas, and 5 other states. For each of those states, Randolph County will have a unique FIPS code (01111 in AL, 37151 in NC). Each FIPS code is

5 digits long, with the first two digits denoting the state (10 denotes Alabama) and the last three digits denoting the county (111 for Randolph county).

countyFIPS contains the FIPS code for a county. These will always be enough to uniquely identify a county, with the exception of FIPS code 0. This references “Statewide Unallocated” in our dataset, meaning that the data is associated with a particular state, but not with a specific county within that state. Any analysis that will take place at a county level will largely ignore this data. However, it may be worth investigating further to ensure it is an insignificant percentage of each state’s total data. If it is the case that a state does not report data at a county level, and a high percentage of its total reported cases are statewide unallocated, then it could skew data analysis at the state level to ignore this data. As a result, in order to uniquely identify rows with a 0 countyFIPS, it is also necessary to take the State into account.

County Name is the name of the county. It is unique within a state, but not necessarily unique between states.

State is the two letter abbreviation for each state. These are unique to each state, and include “DC” for Washington D.C., totalling 51 States.

StateFIPS is the FIPS code for the state the county is located in. In our dataset, StateFIPS range from 1-56. This is because some digits are skipped in the FIPS numbering scheme. For example, Alaska’s FIPS code is 2, followed by Arizona with FIPS code 4, and FIPS code 3 is not used.

County Population is the population of a given county. This data is a single static value, and is not updated over the course of time as are the cases and deaths data.

yyyy_mm_dd_cases keeps track of the cumulative total of cases in a given county by the date denoted in the column title. This means that if the column 2021_01_22_cases reports a value of 100 cases for a given county, and the column 2021_01_23_cases reports a value of 105, 5 total new cases were reported between data collections. Our dataset includes data from January 22, 2020 to July 23, 2023.

yyyy_mm_dd_deaths is the cumulative total of COVID-related deaths in a given county by the date denoted in the column title. According to the data source, a death is counted as a COVID-related death if the virus played a direct role in the death, meaning that the totals in our data do not include everyone who died with COVID-19. As with the cases data, our dataset includes deaths data from January 22, 2020 to July 23, 2023.

Individual Sections - Enrichment Data

Census Demographic ACS - Jesse Carter

The enrichment dataset I chose to examine was the Census Demographic ACS dataset. This dataset includes demographic information about the populations of counties, including breakdowns by age range, sex, and race. This dataset has a large number of variables, many of which pertain to race. Since I am primarily interested in the sex and age demographics, for this stage of the project I will remove the race related variables from the dataset. The variable dictionary below is for the final version of the demographics dataset, after I have trimmed down the variables to only the ones necessary for this stage:

Census Demographic ACS Variable Dictionary

Census Demographic ACS Dataset				
Name	Definition	Data Type	Possible Values	Required
Label (Grouping)	Information about that row of data in the format: "CountyName, State!!DataType" Where DataType can refer to "Estimate" or "Percent"	object (string)	"Baldwin County, Alabama!!Estimate" "San Juan Municipio, Puerto Rico!!Percent"	Yes
Male	Male population of the county as a raw number or percentage.	object (integer in string representation)	"105,437", "55,054", "48.4%", "47.6%"	Yes
Female	Female population of the county as a raw number or percentage.	object (integer in string representation)	"105,437", "55,054", "48.4%", "47.6%"	Yes
Sex ratio (males per 100 females)	Ratio of males per 100 females rounded to the nearest tenth	object (float in string representation)	"93.7", "101.5", "(X)"	Yes
Under 5 years	County population under 5 years of age as raw number or percentage.	object (integer in string representation)	"105,437", "55,054", "48.4%", "47.6%"	Yes
... Age grouping continues in increments of 4 years (5 to 9, 10 to 14, etc)				
85 years and over	County population age 85 and older as raw number or percentage.	object (integer in string representation)	"105,437", "55,054", "48.4%", "47.6%"	Yes
NOTE: The rest of the data related to race demographics has been omitted for this stage.				

Label (Grouping) contains information about the county, state, and data type for the row. The format this is presented in is as follows: "CountyName, State!!DataType" where DataType can be either "Percent" or "Estimate." If the DataType is "Percent," then values in columns Male, Female, and the Age Ranges will be values denoting the percentage of the total population of the county. If the DataType is "Estimate," then values in those columns will be the raw number. For the purpose of merging the data, I will be dropping the "Percent" rows of data, as these values can be calculated from the data if they are needed.

Male and **Female** columns contain either the number of Males and Females within the county, or their percentage of the population.

Sex Ratio (males per 100 females) contains the stated ratio, or an (X) if the row is a Percent data type.

Age Range columns contain either the numerical amount of the population in the given range, or the percentage of total population.

In order to merge this data with the primary COVID-19 dataset, the best option would be to use county and state information found in the "Label (Grouping) variable" to merge with the "State" and "County Name" columns of the COVID-19 dataset. Since this data is embedded in one string variable, it is necessary to parse the strings for that information, and then translate the State Name strings to the State Abbreviations that we have in the COVID-19 dataset. Since county names are not necessarily unique between states, we require both of these pieces of information to merge our data together.

This enrichment data can help in our analysis of COVID-19 by allowing us to compare case and death numbers in counties by their population ages. For example, some initial hypothesis questions could include:

- Are there higher death counts relative to population in counties with a higher proportion of older people?
- Are case numbers higher relative to population in counties with a larger proportion of younger people?

Economic characteristics - Marcus Harrington

Economic characteristics Data set				
Name	Definition	Data Type	Possible values	Required
Label	Information pertaining to that row of data as "County, state!! DataType" where Data type can refer to "estimate", "margin of error", "percent", and "percent margin of error"	Object (string)	"Honolulu County, Hawaii!! Estimate" or "Kauai County, Hawaii!! percent"	Yes
Employment status	The total number of people in various demographics that encompass the workforce	Integer	"167,882", "55.7%", "2.1%", "85,210"	Yes
Commuting to work	Total number of people who use various modes of transportation	Integer	"735", "9,550", "0.8%", "16.7%", "14,589"	Yes
Occupation	Number of People who are employed in various business, construction or, manufacturing occupations	Integer	"34,484", "9,763", "18.8%", "38.6%"	Yes
Industry	Number of people who are employed in economic, STEM, and public industries	Integer	"4,658", "10.1%", "3,743", "16.2%"	Yes
Class of worker	Number of people who fit into government, private, self-employed, or unpaid categories	Integer	"89,349", "21.4%", "0.2%", "143", "67%"	Yes
Income and benefits	People who have made a certain amount of money by households, categories are separated by increments of ~\$10,000	Integer	"7,104", "74,764", "17.3%", "8.7%"	Yes
Health Insurance Coverage	Number of people with public, or private coverage, employed or not	Integer	"112,373", "2,843", "94.3%", "5.7%"	Yes

Label (Grouping) contains information about the country, state and data type for that row. "CountyName, State!!DataType" where DataType can be either the "value", ""Percent" or "margin of error." If the DataType is "Percent," then values in the other Ranges will be values denoting the percentage of the total population of the county. If the DataType is "value," then values in those columns will be the raw number. Lastly, if the DataType is a margin of error then it is listed as a float. For simplicity of merging the data the "Percent" rows will be dropped, as these values can be calculated from the data if they are needed.

Employment status column contains the values of people who are working and those who are unemployed or their percentage of the population with a margin of error

Commuting to work column contains the values of people who commute, carpool, or walk to work as well as the percentage of the population that use these travel methods. It also includes a margin of error

Occupation and Industry columns display the number of people who work in various fields and disciplines represented with a percentage of the population and a margin of error

Class of worker denotes the number of people who fit under a type of employment such as self employed, non employed, or government employed or privately employed. With a percentage and a margin of error

Income and benefits is the number of people who fit into a specific section of income as a percentage or margin of error

In order to merge this data with the primary COVID-19 dataset, the best option would be to use county and state information found in the "Label" variable to merge with the "State" and "County Name" columns of the COVID-19 dataset. Since this data is embedded in one string variable, it is necessary to parse the strings for that information, and then translate the State Name strings to the State Abbreviations that we have in the COVID-19 dataset. Since county names are not necessarily unique between states, we require both of these pieces of information to merge our data together.

This data on economics can prove useful to the overall database by allowing us to compare the changes in the economy over long periods of time. For example some possible hypothesis include

1. Are there higher death counts in people who generate less income than those who make more?
2. Do jobs that require being indoors more often result in higher drops of employment rate?
3. Which group of people who don't have insurance, have private insurance, or have public insurance have more deaths than those who have the other options?

Housing ACS-Collin Harris

The enrichment data set that I chose was the housing data set that includes data on the housing status within each county. This includes Occupancy, Number of Units ,Margin of the Year built , Number of Rooms/BedRooms, etc. The primary focus will be on the housing occupancy area.

Housing Dataset Variable Dictionary

Housing Dataset -04-				
Name	Definition	Data Type	Possible Values	Required
Geography	Information about the data in each column such as "DataType!!Category!!SubCategory" Dtype is either an "Estimate", "Margin of Error" or "Percent" with Annotations that will be omitted Category would be the specific categorical quantity the data is trying to measure In this case Housing Occupancy below are the listed sub categories I am focusing on.	object(string)	"Estimate!!HOUSING OCCUPANCY!!Total housing units" ,"Margin of Error!!HOUSING OCCUPANCY!!Total housing units"	Yes
Occupied Housing Unit	The number of houses in a county that are currently being occupied, represented as a raw number, percentage, and margin of error for both.	object	0, 14,415,594 , ±1,000, 72.3% ±0.1,(X)	Yes
Vacant Housing Units	The number of houses in a county that are currently not being occupied, represented as a raw number, percentage, and margin of error for both.	object	0, 14,415,594 , ±1,000, 72.3% ±0.1,(X)	Yes
Homeowner Vacancy Rate	The rate at which houses are being vacated	object	7.1, ±5.3, (X)	Yes
Rental Vacancy Rate	The rate at which houses are being vacated	object	4.1, ±0.9,(X)	Yes
... More data on: Unit Structure, Unit Year Built, Rooms, Bedrooms, Housing Tenure, The Year Current Homeowner Moved In, Vehicle Availability, House Heating Fuel, Selected Characteristics, Occupants				

per Room, Value of the House, Mortgage Status, Selected Monthly Owner Costs, Selected Monthly Owner Costs as a Percentage of Household Income (SMOCAPI), Gross Rent, and Gross Rent as a Percentage.

Housing

Geography-includes the geographical id of all counties included in the dataset in its column, in its row are the labels that follow the syntax of “DataType!!Category!!SubCategory” as my primary focus is Housing the syntax will look like “DataType!!HousingOccupancy!!SubCategory” for reference.

Occupied Housing Unit-numerical quantity of houses that are being occupied at the time of the census

Vacant Housing Units-numerical quantity of houses that are not being occupied at the time of the census

Homeowner Vacancy Rate-the rate at which homeowners are vacating their own property.

Rental Vacancy Rate- the rate at which rental properties are being vacated at. Our final merged data of all the covid data and this housing data set would be best merged with “GEO_ID” which contains the “FIPS” for each county as well as the county names. Covid had a huge impact on people's home lives this enrichment data set can help us generate questions such as:

- What was the change in occupancy rate due to covid cases and deaths?
- Would the change in occupancy be greater within rental properties, such as apartments, townhomes and condos, rather than owned properties?

Quarterly Census of Employment and Wages - Emanuel Macias

The enrichment dataset chosen was the quarterly census of employment and wages(QCEW). As the name suggests it gives information about the earnings of each state and country. The data includes quarterly earnings as well as employment based on the first three months of the year, and average weekly earnings. Some other data that is included in the data set is the sector or ownership from the earning. Whether they are from a private sector or from the government. It includes local, state and federal government earnings from each county. Some private industries included are manufacturing, construction, education and health services. There is also some data that is relevant for this part of the project. This includes data like quotients relative to the US, the monthly employment and average weekly wage. The following variable dictionary below is for the final version of the demographics dataset, after some irrelevant variables were removed:

Quarterly Census of Employment and Wages:

QCEW Dataset				
Name	Definition	Data Type	Possible Values	Required
Area Code	These are unique to each county. Also known as FIPS	int64 (integer)	0, 1001, 56045	Yes
County Name	Name of the County. These are not necessarily unique.	object (string)	Weston County, Yakutat Borough, Yukon-Koyukuk Census Area	Yes
St Name	State Name	object (string)	AL, NC, WY	Yes
Ownership	The source of the data or wages.	object (string)	Federal, Local, Private	Yes
Industry	The type of work the wages are coming from	Object (string)	Information, Construction, Manufacturing	Yes
Total Quarterly Wages	The total earning of the country per quarter of the year	int64 (integer)	0, 8149, 70521	Yes
Establishment Count	The number of business during the period of time	int64 (integer)	0, 8149, 70521	Yes
NOTE: The rest of the data for QCEW has been omitted for this stage.				

The listed rows in the variable dictionary give information on what the relevant data needed for this stage of the project. **Area code** is equal to the county FIPS number, which

uniquely identifies each county. As mentioned, the county name sometimes repeats in different states. Each state has a unique county name, but those county names may be repeated in different states. Another important fact to consider is the variable **Ownership**. The data set has different values as shown in the dictionary. Though focusing on the total covered value in the ownership variable would be most relevant. As this gives a comprehensive look at employment and wages for each county.

To merge the data set with the COVID-19 data the area code would be the best way to do this. As it corresponds to the FIPS county code, which uniquely identifies each county. The state and county name would also be helpful to identify uniqueness. Using the **total covered** value for the **Ownership** variable would be best to eliminate extraneous values for each county.

The enrichment data would prove useful when being compared to the COVID-19 dataset. It can be used to identify any changes in employment when the covid 19 was at its peak or during county outbreaks. Another potential use case would be whether the wages had some correlation with the outbreak. A hypothesis would be if counties with higher wages suffered less death. If higher population and cases resulted in lowered establishment count.

US Election 2020 - Sumithra Hariguruprasad

Introduction:

The US election 2020 was a historic and closely watched presidential election held on Tuesday, November 3, 2020. It featured a contest between the former Vice President Joe Biden, the Democratic candidate and the incumbent President Donald Trump, the Republican candidate. This report provides an overview of the data contained in the "president_county_candidate.csv" dataset, along with a variable dictionary and data types for each variable.

DataSource:

The data used for this analysis is sourced from the Kaggle (www.kaggle.com), which contains information about the presidential election results at the county level, including state, county, candidate name, candidate party name, total votes, and whether the candidate won in that county.

Variable Dictionary and Data Types:

president_county_candidate				
Name	Definition	Data Type	Possible Values	Required
state	State name	object	Deleware, Indiana, Missouri	Yes
county	County name	object	Kent County, Brown County, Andrew County	Yes
candidate	Candidate name	object	Joe Biden, Donald trump, Howie Hawkins	Yes
party	Candidate party name	object	DEM, REO, IPO, LIB	Yes
total_votes	Total votes	int64	0 - 3.03 million	Yes
won	Won race	bool	True/False	Yes

- **state** (Data Type: object) - Represents the name of the state where the county is located.
- **county** (Data Type: object) - Represents the name of the county within the state.
- **Indidate** (Data Type: object) - Indicates the name of the presidential candidate.
- **party** (Data Type: object) - Denotes the political party affiliation of the candidate.
- **total votes** (Data Type: int64)- Represents the count of votes received by the candidate in the county.
- **won**(Data Type: Bool) -Indicates whether the candidate won in the county or not (True/False).

Data Types Explanation:

- **object**: Data represented as alphanumeric characters, used for state names, county names, candidate names, and party names.
- **int64**: Data represented as integer values, typically for counts, percentages, or amounts, used for total votes.
- **Bool**: Data with two possible values, typically representing true or false, used to indicate whether a candidate won in a county.

Conclusion:

The dataset 'president_county_candidate.csv' provides information about the US Election 2020 results at the county level, including the candidates, their party affiliations, and the number of votes they received in each county. Using the Pandas merge function this dataset is combined with covid datasets based on common variables. In this case 'state' is used as the mapping variable. The combined dataset can be of help in the analysis of COVID-19 spread. Some of the initial hypothesis questions I plan to investigate include::

- Is there a correlation between voter turnout and the number of covid cases?
- Do the counties with different political affiliations (e.g., Democratic, Republican) exhibit variations in the rate of COVID-19 spread?